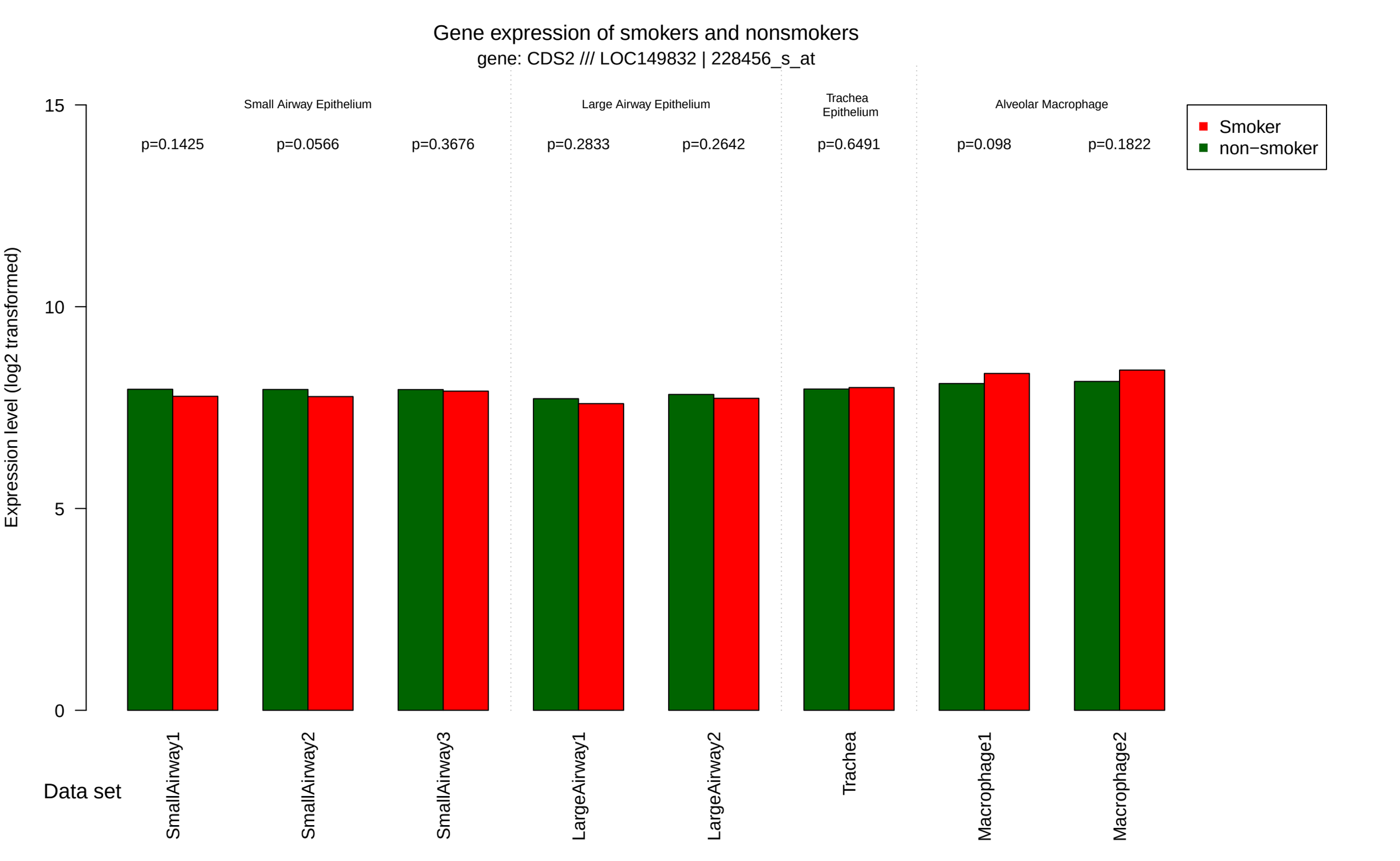
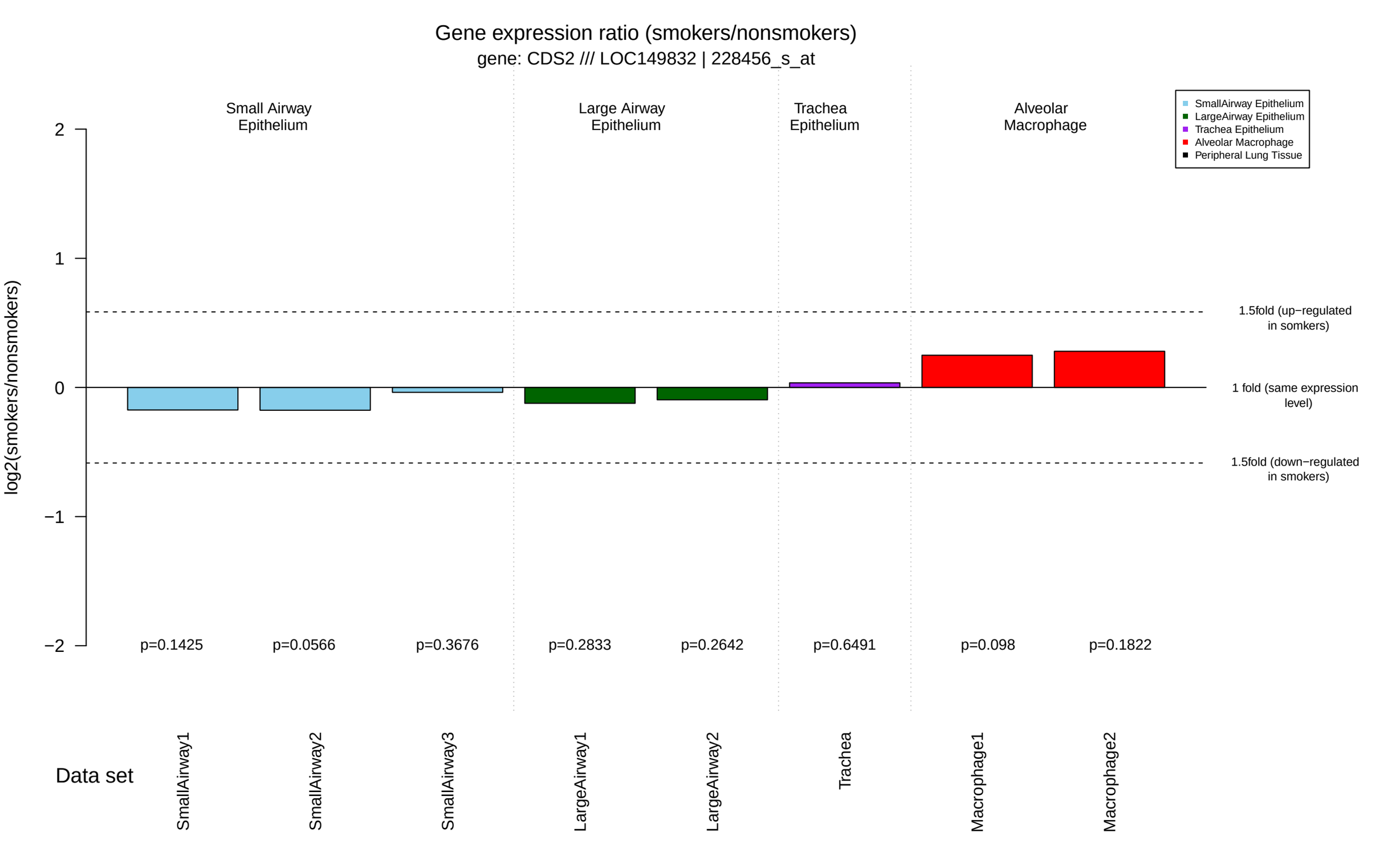
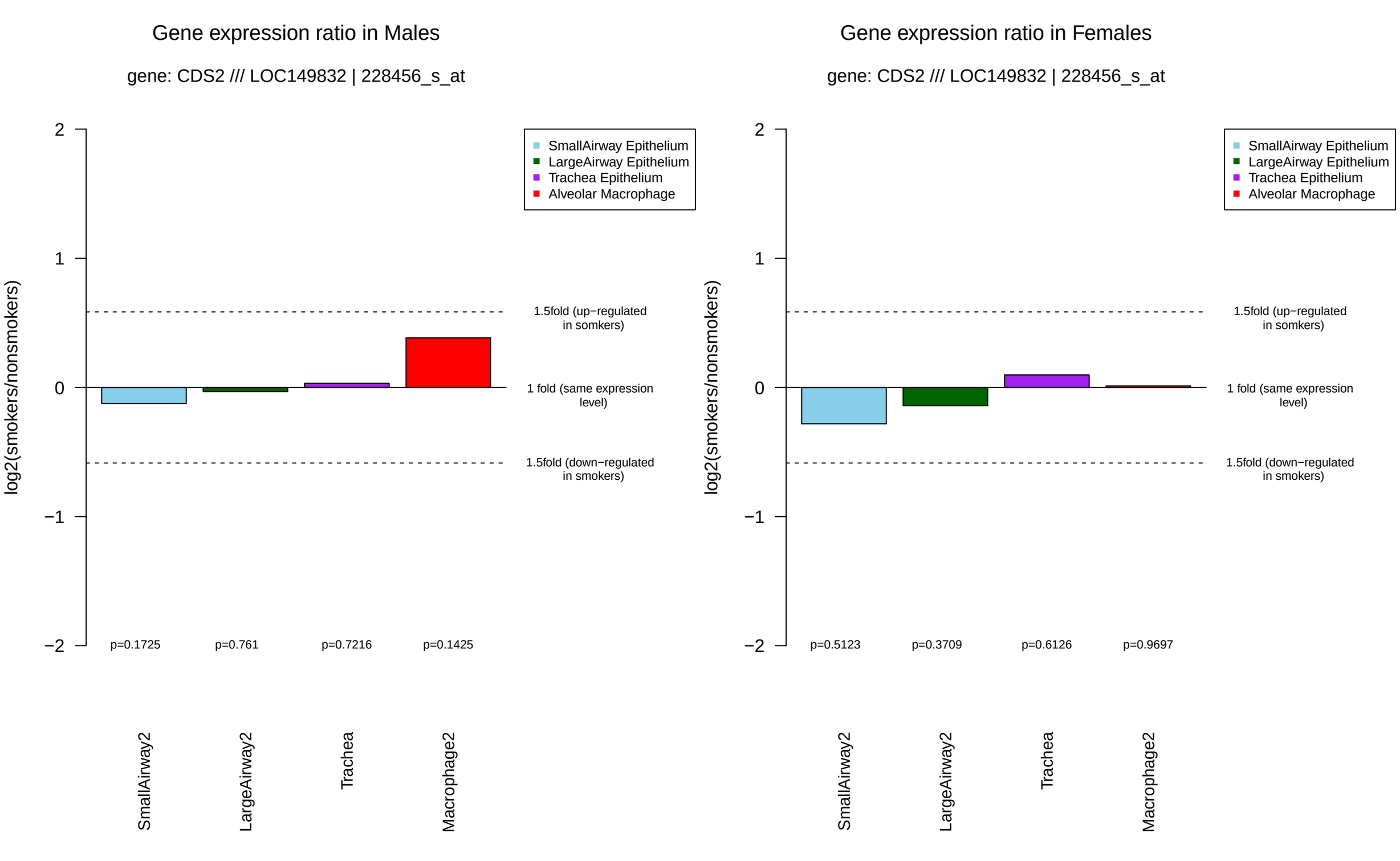
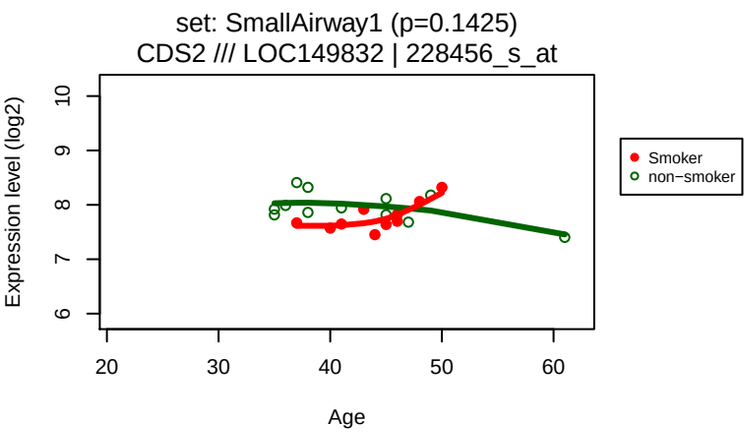
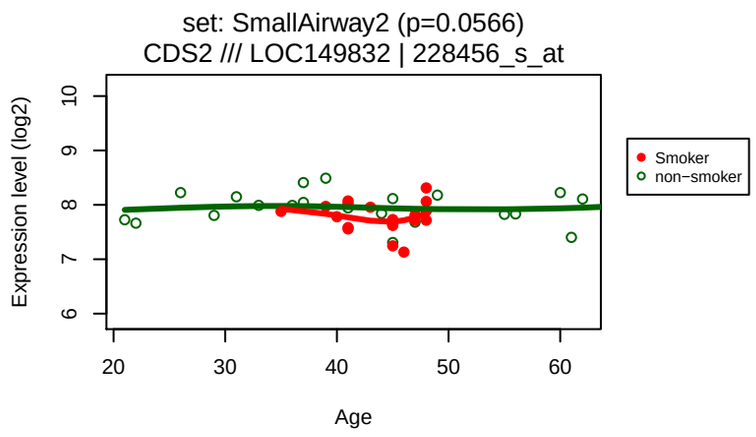
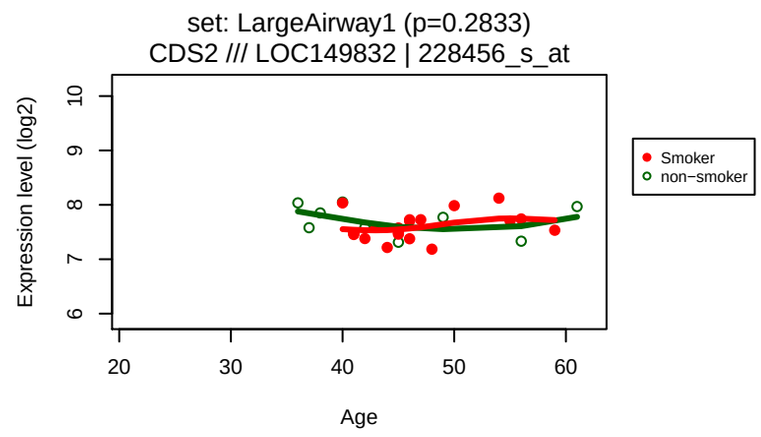

Gene expression of smokers and nonsmokers
gene: CDS2 /// LOC149832 | 228456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1425
p=0.0566
p=0.3676
p=0.2833
p=0.2642
p=0.6491
p=0.098
p=0.1822
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

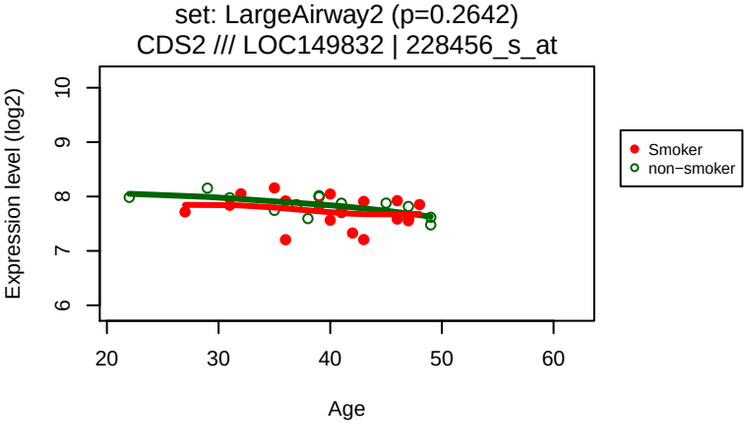

Gene expression ratio (smokers/nonsmokers)
gene: CDS2 /// LOC149832 | 228456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1425
p=0.0566
p=0.3676
p=0.2833
p=0.2642
p=0.6491
p=0.098
p=0.1822
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

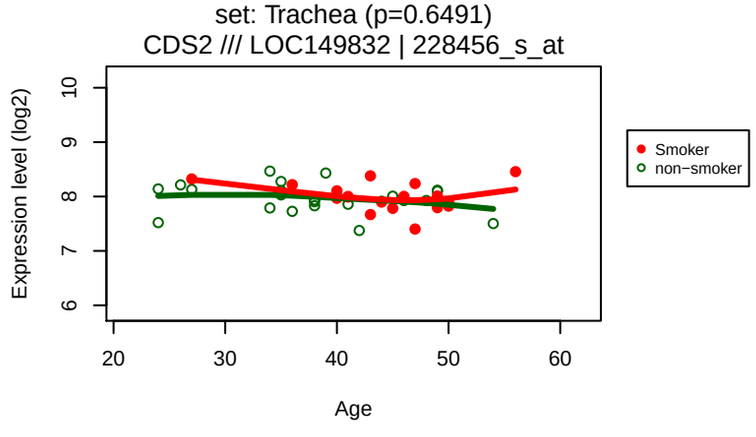

Gene expression ratio in Males
gene: CDS2 /// LOC149832 | 228456_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1725
p=0.761
p=0.7216
p=0.1425
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDS2 /// LOC149832 | 228456_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5123
p=0.3709
p=0.6126
p=0.9697
SmallAirway2
LargeAirway2
Trachea
Macrophage2

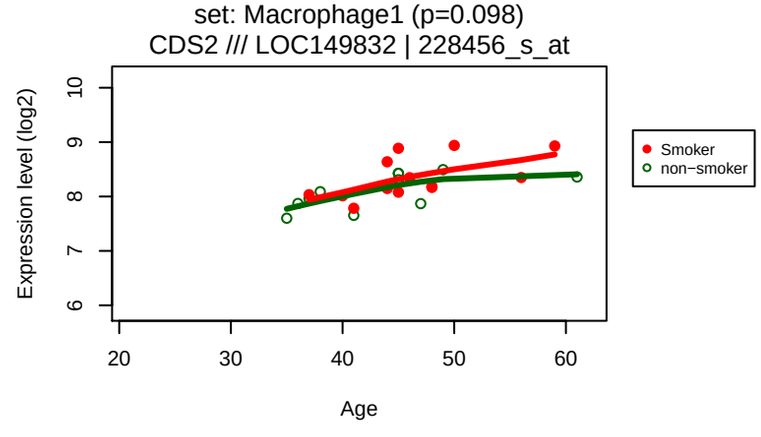

set: SmallAirway1 (p=0.1425)
CDS2 /// LOC149832 | 228456_s_at
Expression level (log2)
Age
Smoker
non−smoker

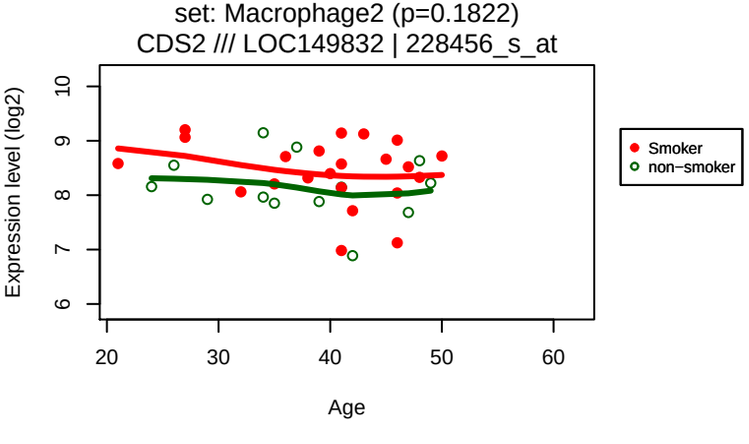

set: SmallAirway2 (p=0.0566)
CDS2 /// LOC149832 | 228456_s_at
Expression level (log2)
Age
Smoker
non−smoker

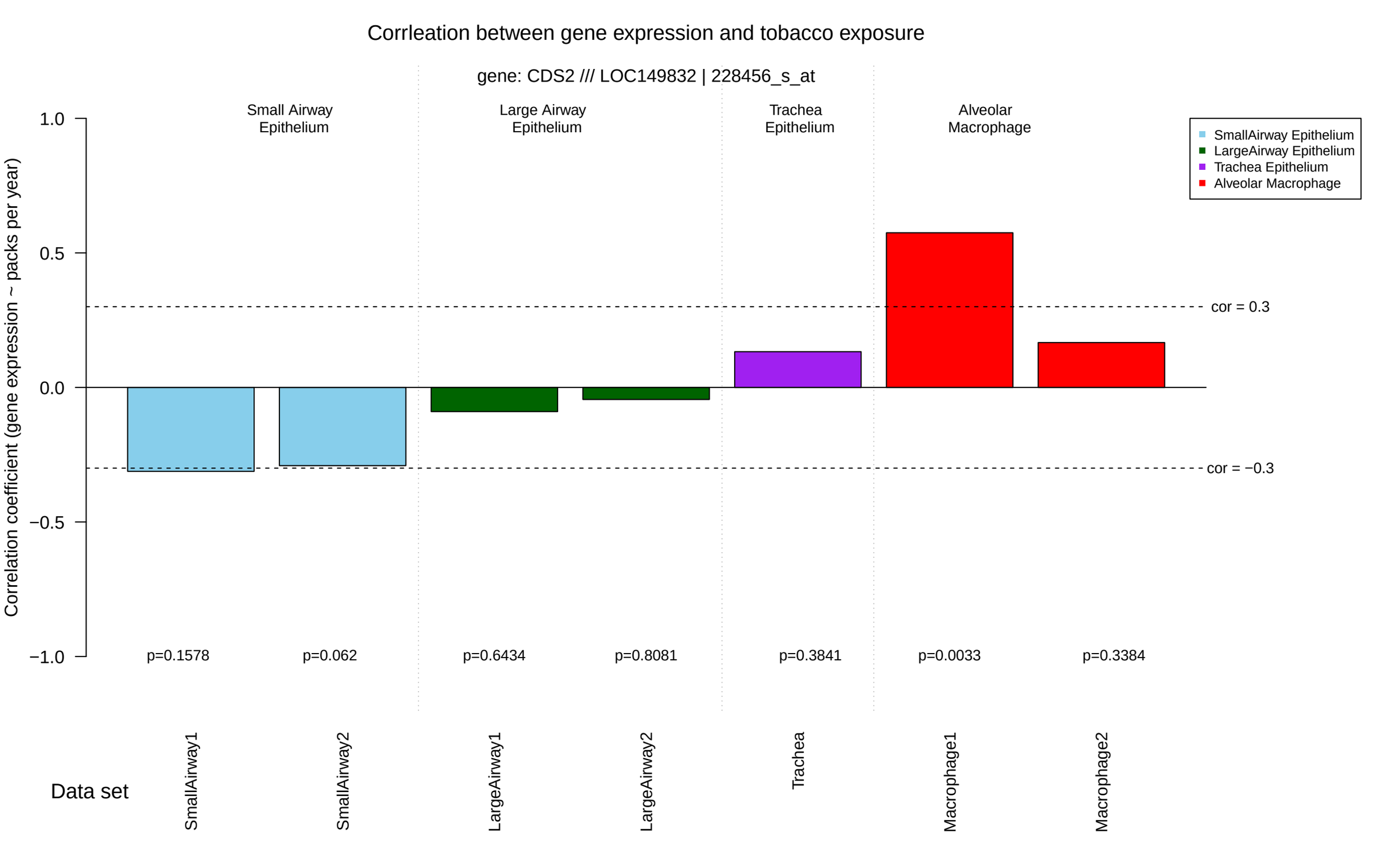

set: LargeAirway1 (p=0.2833)
CDS2 /// LOC149832 | 228456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2642)
CDS2 /// LOC149832 | 228456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6491)
CDS2 /// LOC149832 | 228456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.098)
CDS2 /// LOC149832 | 228456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1822)
CDS2 /// LOC149832 | 228456_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDS2 /// LOC149832 | 228456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1578
p=0.062
p=0.6434
p=0.8081
p=0.3841
p=0.0033
p=0.3384
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2